
IAL Physics Unit 3 (WPH13): A* Revision Guide

Practical Skills in Physics I

Compiled from 21 WPH13 papers and their mark schemes (June 2019 – June 2026). Mark-scheme wording is quoted where the exact phrase earns the mark.

1. The paper and where the marks are

Format. 1 hour 20 minutes, 50 marks, 4 structured questions (about 10–20 marks each). There is no multiple choice and no 6-mark extended question. Every question is a practical set in a Unit 1 or Unit 2 context, and the same **skills** are tested every time. Learn the skills, and the physics contexts follow from Units 1 and 2.

Skill	How often	Typical marks and form
Uncertainty of a reading / mean / percentage	Every paper (2–3 times)	Half resolution; mean of repeats; half range; % uncertainty; 1–3 marks each.
Comparing a result with an accepted value	Every paper	Uncertainty-range limit or percentage difference, then a conclusion; 2–3 marks.
Graph plotting	Every paper	Calculate a new column to 3 s.f., label axes, choose scales, plot, draw the line: 4–6 marks.
Linearising an equation	Every paper	Compare with $y = mx + c$; state what the gradient/intercept equals; 2 marks.
Gradient → physical quantity	Every paper	Large triangle, 2–3 s.f., unit; 3–4 marks.
Criticising a results table	Almost every paper	2–3 faults (d.p./s.f., repeats, number of readings, anomaly).
Planning / method / technique	Every paper	Instrument + how to use it + how to reduce error; 3–5 marks.
Safety, control variables, circuits	Most papers	Hazard + control measure (2 marks); circuit diagram (2 marks); variable + how it is controlled.

HOW THESE MARKS ARE GIVEN

1. Most 2-mark “explain” items are **linked pairs**: an action plus its reason (“check for zero error *to eliminate systematic error*”). The second mark depends on the first. Always write both halves.
2. Quote values to the precision asked: a mean to the same number of decimal places as the data; a calculated column to 3 s.f.; a gradient-derived constant to 2 or 3 s.f. **with its unit**; a percentage uncertainty to 1 or 2 s.f.
3. Bare answers in data analysis often score nothing; show the substitution.
4. A conclusion mark needs a **numerical comparison** and a decision (“ $8.73 < 8.94$, so the prism is not copper”).

2. Instruments and their resolutions

Instrument	Resolution	Use and reason to choose it
Metre rule / tape measure	1 mm	Lengths above about 10 cm. Keep it vertical (set square or spirit level), close to the object, and read at eye level, perpendicular to the scale.
Vernier calliper	0.1 mm	Diameters/thicknesses of a few cm; no parallax; tips fit curved surfaces. Better resolution than a rule, so smaller % uncertainty.
Digital calliper	0.01 mm	As vernier; also internal/external diameters. Check the zero.
Micrometer screw gauge	0.01 mm	Wire diameter, paper/coin thickness, ball bearings. Use the ratchet (prevents over-tightening, which deforms the object or can break glass).
Protractor	1° (0.5° on some)	Angles; align the zero with the normal; a spectrometer-type scale reads to 0.1°.
Stopwatch	0.01 s	But human reaction time (about 0.2 s) dominates. Time many oscillations or a long interval, or use light gates, a data logger or video.
Balance	1 g, 0.1 g or 0.01 g	Zero (tare) first. Measure many items together (20 sheets, 20 coins) and divide.
Thermometer	1 °C (digital 0.1 °C)	Digital: higher resolution, no parallax. Stir the water; put the thermometer next to the sample.
Multimeter / digital meters	Last digit (e.g. 0.01 V)	No parallax, no interpolation, range can be changed for higher resolution.
Oscilloscope (CRO)	Time per division setting	Time between pulses = divisions $\times$ time base; measure over several divisions or cycles.
Light gates + data logger	About 0.001 s	Removes reaction time and records simultaneously; but must be aligned at the right height and needs extra set-up.

3. Uncertainties: the rules the mark schemes use

Single reading	Uncertainty = half the resolution (micrometer 0.005 mm; metre rule 0.5 mm; balance 0.5 g). Using the full resolution usually scores 1 of the 2 marks. A <i>length</i> from a rule involves two readings, but a single “ $\pm$ half resolution” is what the scheme normally expects unless told otherwise.
Repeated readings	Mean of all values (or exclude a clear anomaly and say so). Uncertainty = half the range (or the largest difference from the mean). Using the whole range scores nothing.
Percentage uncertainty	% U = (uncertainty $\div$ value) $\times$ 100. Example: (0.005 mm $\div$ 1.72 mm) $\times$ 100 = 0.29 %.
Multiple items	Measuring the thickness or mass of 20 sheets/coins together gives a larger reading with the same absolute uncertainty, so the % uncertainty is smaller. Divide the result by 20.
Larger quantity, same instrument	The absolute uncertainty is the same, the reading is larger, so the % uncertainty is smaller. Examples: a longer wire, a larger separation of maxima, a greater distance to the screen, several wavelengths or periods.
Combining (asked occasionally)	% U adds when multiplying or dividing; a power multiplies it (% U in $r^3 = 3 \times$ % U in r ; in $A = \pi d^2/4$ it is $2 \times$ % U in d).

COMPARING A RESULT WITH AN ACCEPTED VALUE (2-3 MARKS, EVERY PAPER)

Method A, uncertainty limit: take your value and the stated % uncertainty; calculate the relevant limit (value $\times$ (1 $\pm$ %)); state whether the accepted value lies inside the range.

Method B, percentage difference: % difference = $|\text{accepted} - \text{yours}| \div \text{accepted} \times 100$ (the denominator must be the published value); compare with the stated % uncertainty and conclude.

Example (Jun 2025): $\rho = 8.65 \text{ g cm}^{-3}$ with 0.9 % uncertainty. Upper limit = $8.65 \times 1.009 = 8.73 \text{ g cm}^{-3} < 8.94 \text{ g cm}^{-3}$ (copper), so the prism is not copper. Or: % difference = $(8.94 - 8.65)/8.94 = 3.2 \% > 0.9 \%$, so not copper. Calculating the limit of the *accepted* value instead of your own scored zero in Jan 2026.

Random and systematic error

Random error	Causes scatter in repeated readings (reaction time, parallax, judging a maximum or minimum, temperature fluctuations). Reduced by repeating and taking a mean, by timing longer intervals, or by a line of best fit through many points. It cannot be eliminated.
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Systematic error	Shifts every reading the same way (zero error, a stretched or warped rule, a meter that reads high, unblocked background light). Repeating does not remove it. Remove it by checking and correcting the zero, calibrating, or controlling the variable. On a graph it appears as an unexpected intercept, so the gradient method “ignores” it.
Precision vs accuracy	Precise: repeats close together (small spread). Accurate: close to the true value.
Why a graph beats one calculation	It uses more pairs of values, the best-fit line averages them, anomalies stand out, and the gradient is unaffected by a systematic (zero) error.

4. Results tables: what examiners criticise

“Criticise the table” (2–3 marks) is in almost every paper. Choose from this list, each stated specifically.

- Inconsistent number of **decimal places** in a measured column (e.g. 18.4, 19, 19.25).
- Inconsistent **significant figures** in a calculated column.
- Values not recorded to the instrument’s resolution (rule → nearest mm).
- **No repeat readings** shown (do not write this if repeats are in the table).
- **Too few readings** (fewer than 5–6) to plot a reliable graph.
- **Range too small**; no values in an important region.
- Uneven intervals or a large gap in the independent variable.
- A value that **does not follow the trend** (identify it: “0.74 V does not fit”); or a calculated value that is wrong or incorrectly rounded (“9.9 should be 9.96”).
- Missing or incorrect units in a column heading (heading format: quantity / unit).

5. Graphs

THE PLOTTING MARKS (4 MARKS PLUS 1-2 FOR A NEW COLUMN)

1. **New column:** calculate every value correctly, to 3 s.f. (e.g. $\sin \theta$, $1/\lambda$, d^2 , $1/t^2$, v^2 , R/L), with a heading in the form “ $1/\lambda / 10^6 \text{ m}^{-1}$ ”.
2. **Axes labelled** “quantity / unit” the correct way round (the variable named first on the y-axis, e.g. “plot λ against $\sin \theta$ ”). Use no unit for $\sin \theta$ or n ; write $\sin(\theta/^\circ)$, not $\sin \theta / ^\circ$.
3. **Scales:** 1, 2 or 5 units per 2 cm square (never 3 or 7); plotted points covering **at least half** the grid in both directions. Axes need not start at zero.
4. **Plotting:** every point within 1 mm (half a small square); one error loses the mark (two plotting marks when there are many points).
5. **Line of best fit:** a single thin straight line (or smooth curve if the trend is curved) with points balanced on both sides; do not force it through the origin or join the dots.

- **Gradient:** draw a large triangle covering more than half the line; read values **from the line** (table points only if they lie on it); show the working; give 2-3 s.f. with units.
Reading a single point and dividing (unless the line passes through the origin) scores nothing.
- **Intercept:** read it where the line crosses $x = 0$ (check the axis really starts at 0), or calculate $c = y - mx$ from a point on the line.
- **Proportionality test:** a straight line **through the origin** shows $y \propto x$. Two statements (straight line, through origin) plus the conclusion. For inverse proportion, plot y against $1/x$, or check that xy is constant for several pairs.
- **Finding a maximum (Jan 2025, solar panel angle):** draw a smooth curve, read the peak, and improve by taking more readings at smaller intervals around the peak.

Linearisation bank (from these papers)

Equation	Plot (y against x)	Gradient / intercept
$n\lambda = d \sin \theta$ (electron diffraction, $n = 1$)	λ against $\sin \theta$	gradient = d
$n_1 \sin \theta_1 = \sin \theta_2$	$\sin \theta_2$ against $\sin \theta_1$	gradient = n_1
$eV = hc/\lambda - \Phi$ (or $V_a = (hc/e)(1/\lambda) + W/e$ for LEDs)	V (or E) against $1/\lambda$	gradient = hc/e (or hc); $h = \text{gradient} \times e/c$
$\lambda = h/(mv)$	λ against $1/v$	gradient = h/m
$I = k/d^2$	I against $1/d^2$	gradient = k , through the origin
$R/L = -kL + \rho/A$	R/L against L	intercept $\times A = \rho$
$R = \rho l/A$ (track: $l = 2x + 0.05$)	R against x	gradient = $2\rho/A$
$R = R_0(1 + \alpha\theta)$	R against θ	intercept = R_0 ; $\alpha = \text{gradient}/R_0$
$R = \rho l/(wt)$	R against $1/w$ (or l)	gradient = $\rho l/t$
$F = \mu N$ (friction)	F against N	gradient = μ (no unit)
$d = (l^3/4wh^3E)F$ (beam)	d against F	$E = l^3/(4wh^3 \times \text{gradient})$
$r = e^2h$ (bounce)	r against h	gradient = e^2
$D^2 = (2v^2/g)H$ (projectile range)	D^2 against H	gradient = $2v^2/g$
$\cos(\theta/2) = (mg/k)(1/\Delta x)$	$\cos(\theta/2)$ against $1/\Delta x$	gradient = mg/k
$1/t^2 = (g/3s^2)h - \dots$	$1/t^2$ against h	$g = \text{gradient} \times 3s^2$
$f = (v/2L)n + k$	f against n	$v = 2L \times \text{gradient}$
$f = (v/\lambda) \sin \theta$	f against $\sin \theta$	$v = \text{gradient} \times \lambda$
$F = (M/t)v$ (impulse)	F against v	gradient = M/t
$m = (C\rho A/2g)v^2$ (drag)	m against v^2	$C = \text{gradient} \times 2g/(\rho A)$
$v^2 = kd$ (water waves)	v^2 against d	gradient = k ($\approx g$)

$l = (\text{efficiency} \times VI/mg)t$	l against t	gradient = speed
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Write the linearising answer as: rearrange the equation; write it under $y = mx + c$, matching each term; state “so the gradient = ... (and ... are constants)”. That is 2 marks.

6. Techniques that earn the method marks

Parallax	Use a set square between the scale and the object; read at eye level / perpendicular to the scale; put the rule close to (touching) the object; attach a pointer/marker; clamp the rule.
Vertical / horizontal	Set square against the bench or a spirit level/plumb line; measure the height in two places.
Zero error	Close the callipers/micrometer, or tare the balance, and check the reading is zero; correct readings if not. This eliminates systematic error.
Diameter / thickness	Micrometer (with ratchet) at several places along it and in different orientations; take the mean; check the zero. Measure at the widest point of a sphere.
Timing	Time several oscillations or a long distance; repeat and average; start and stop as the same point passes a marker, viewing at eye level; use light gates, a data logger or a video (slow motion, frame-by-frame) to remove reaction time.
Terminal velocity check	Two (or more) equal sections between rubber bands: equal times mean constant velocity. Start the first band far enough below the surface.
Resonance / maxima	Approach the frequency from above and below and take the mean; detect the maximum with a microphone and CRO or a sound level meter rather than the ear.
Light experiments	Darken the room or cover the apparatus to block background light (or measure and subtract it); keep the distance and angle fixed; use monochromatic light (laser, LED) instead of filters.
Electrical	Fixed or protective resistor to limit current (so no overheating or short circuit, and no damage to the ammeter or supply); start a variable resistor at maximum; switch off between readings to stop heating; minimise contact resistance by cleaning the contacts.
Heating / cooling	Water bath (kettle or Bunsen) to heat, ice bath to cool; stir; thermometer close to the sample; wait for a steady temperature.

Safety: hazard + control (always as a pair)

Hazard	Control measure
Masses falling (on feet)	Keep feet clear; box/cushion/sand tray underneath; wear safety boots.
Wire, cord, rubber band or nylon snapping into the eyes	Wear safety goggles (not just “glasses”); use a safety screen.

Stand toppling	Clamp the stand to the bench, or put a heavy mass on its base.
Laser light damaging the eye	Never look into the beam; stand behind the laser; avoid reflective surfaces; laser safety glasses.
Hot water / hot apparatus (scalds, burns)	Clamp the beaker; use heat-proof gloves; let it cool before handling.
Electric shock near water	Low-voltage supply; keep the supply away from the water bath.
Loud sound damaging hearing	Ear defenders; limit the volume and the time of exposure; do not stand close to the speakers.
Broken glass (micrometer on a tube, glass beam)	Use the ratchet; wear gloves and eye protection.
Higher-energy (UV) light in photoelectric work	Use visible light/LEDs where possible; do not look at the UV source.

7. The practicals: methods examiners expect

Each entry gives the measurements, the key technique details and the analysis, as rewarded in these mark schemes.

Density (sphere, cube, coins, paper, prism)	Mass on a balance (many items together). Dimensions with callipers/micrometer at several positions, mean (exclude an anomaly). $V = \frac{4}{3}\pi r^3$, $l \times w \times h$, $\pi r^2 t$, or $\frac{1}{2}ab \times c$ for a prism. $\rho = m/V$ to 3 s.f., then compare with the table using a % range. Irregular coins: displacement can or measuring cylinder with many coins, then divide.
Spring constant / Hooke's law	$F = mg$ (or a newtonmeter); extension = loaded length – original length (or align the zero with the end); pointer and set square to avoid parallax; range of loads; k = gradient of the linear section of the F - x graph.
Young modulus / breaking stress	Long thin wire, clamp, marker, scale; micrometer diameter (mean of several); $F = mg$; stress-strain gradient. For breaking stress, add masses gradually until it breaks, smaller increments near breaking, and repeat with new samples. Goggles; keep feet clear. Area scaling: 1/20 of the diameter gives 1/400 of the area.
Energy stored (rubber band)	Load and unload, plot force against extension; energy = area (count squares $\times$ energy per square); the area between loading and unloading curves is the energy transferred to thermal energy.
Viscosity (falling ball / pulled sphere)	Rubber bands far enough below the surface; equal sections show terminal velocity; v = distance/mean time; $D = W - U$ (or the difference of two force-meter readings); $\eta = D/6\pi r v$. The temperature must be constant. The largest uncertainty is usually the short time (reaction time 0.2 s is a big fraction) or the force-meter resolution.

g / acceleration	Light gate with a card of measured length: $v = \text{length}/\text{time}$; drop distance with a metre rule; $v^2 = 2as$; repeat and average. Or plot s against t^2 . Moving masses between the hanger and the glider keeps the total mass constant (Newton's 2nd law test).
Momentum (gliders)	Light gates and data logger for the velocities (no reaction time); $p = mv$ to 3 s.f.; compare the % difference of totals before and after. Pushing by hand gives a different force each time, so there is more spread.
Speed of sound	Resonance tube: closed end, $\lambda = 4L$, mean of repeats, $v = f\lambda$; it is hard to judge the loudest point. Echo: distance with a tape measure or trundle wheel, $v = 2d/t$; use a CRO or microphones instead of a stopwatch. CRO: $t = \text{divisions} \times \text{time base}$; check that both pulses fit on the screen.
Stationary waves on a string	Length to the pulley with a rule held against the string; find resonance from above and below; $\mu = mg/(f\lambda)^2$; zero the balance.
Two-source interference	Speakers on the same signal generator (coherent); reversing one speaker's connections puts them in antiphase; mean fringe spacing over several gaps; $\lambda = ws/D$; ear defenders. Microwaves: $\lambda = 2 \times \text{mean distance between maxima}$.
Refractive index	Semicircular or rectangular block on paper; mark the outline and the centre; draw the normal with a set square or protractor; measure θ_1, θ_2 for at least 5 angles (both sides of the normal); plot $\sin \theta_2$ against $\sin \theta_1$; gradient = n . Critical angle: $\sin C = 1/n$. Use monochromatic light (different colours refract differently) and a narrow beam.
Diffraction (grating, hair, electrons)	Measure across several maxima/minima and divide; increase the screen distance or use a longer λ to spread the pattern; laser safety. Electron rings: measure the diameter to the middle of the ring in several orientations (flexible tape or callipers on the curved tube); the rings show interference, a wave property.
Resistivity	Micrometer diameter (mean); length between contacts; V and I (or an ohmmeter, checking its zero); plot R against l ; $\rho = \text{gradient} \times A$. Pencil tracks and strips: $A = \text{width} \times \text{thickness}$. An intercept at $l = 0$ comes from contact or lead resistance.
e.m.f. and internal resistance	Variable resistor (start at maximum) and protective resistor; voltmeter across the cell, ammeter in series; plot V against I ; intercept = ε , gradient = $-r$. Readings change little on repeating, but temperature changes cause drift.
I-V / sensors	Ammeter in series, voltmeter in parallel with the component, potential divider or variable resistor; diode/LED needs a protective resistor; diode temperature controlled with a water bath and thermometer; LDR: fixed distance, background light blocked, distance measured with a metre rule.

Planck constant	LEDs: threshold p.d. V_a for each colour; plot V_a against $1/\lambda$; $h = \text{gradient} \times e/c$. An LED emits a band of wavelengths, so there is uncertainty in λ and h; take more readings near the threshold. Photocell: stopping p.d. for each filter; $E = eV$; plot E against $1/\lambda$. Filters pass a band of wavelengths (systematic error), so use LEDs or lasers; it is hard to judge exactly when $I = 0$, so use a more sensitive ammeter.
Solar cell / motor efficiency	Output $P = VI$ (voltmeter across the cell or motor, ammeter in series); input = $I \times \text{area}$ (or VI for a motor); useful output = mgh/t ; efficiency = output/input. Control the lamp distance, angle and background light; use a lens to give a parallel beam. Data logger: distance and time recorded simultaneously at a high sampling rate.
Inverse square law (light)	Measure to the filament centre (add the bulb radius); set square for parallax; dark room or subtract the background; plot I against $1/d^2$.

8. Fully worked answers to the most common items

UNCERTAINTY OF A SINGLE READING (JUN 2023). Micrometer reads 1.72 mm; resolution 0.01 mm, so uncertainty 0.005 mm; % U = $0.005/1.72 \times 100 = 0.29\%$.

MEAN AND % UNCERTAINTY FROM REPEATS (JUN 2024). $v = 335, 347, 339, 342 \text{ m s}^{-1}$; mean = 341 m s^{-1} (3 s.f.); half range = $(347 - 335)/2 = 6 \text{ m s}^{-1}$; % U = $6/341 \times 100 = 1.8\%$.

GRADIENT TO A CONSTANT (OCT 2023). $E_{\text{max}} = hc(1/\lambda) - \Phi$; gradient = $(12.8 - 6.6) \times 10^{-20} \text{ J} / (2.2 - 1.7) \times 10^6 \text{ m}^{-1} = 1.24 \times 10^{-25} \text{ J m}$; $h = 1.24 \times 10^{-25} / 3.00 \times 10^8 = 4.1 \times 10^{-34} \text{ J s}$; % difference from 6.63×10^{-34} is 38 %, so the method has a large systematic error (filters pass a band of wavelengths).

USING AN INTERCEPT (JUN 2024). R/L against L : intercept $20.9 \Omega \text{ m}^{-1}$; $A = \pi(0.27 \times 10^{-3})^2/4 = 5.73 \times 10^{-8} \text{ m}^2$; $\rho = 20.9 \times 5.73 \times 10^{-8} = 1.2 \times 10^{-6} \Omega \text{ m}$.

ENERGY FROM A GRAPH AREA (JUN 2026). One square = 0.01 J; about 19 squares under the force-extension curve, so energy stored = 0.19 J.

9. Phrase bank for 1-2 mark explanations

Why repeat readings?	Readings vary (random error, e.g. reaction time), so repeating allows a mean to be calculated, which reduces the effect of random error. Repeats also let you identify and remove anomalies.
Why are repeats not needed?	Repeat readings would show little or no variation (random error negligible), e.g. stable digital meter readings.
Why use light gates?	They avoid human reaction time, so they reduce random error/uncertainty in time; but they must be aligned and need a data logger.

Why the vernier rather than the rule?	Smaller resolution (0.1 mm vs 1 mm), so a smaller percentage uncertainty; no parallax; the tips fit curved surfaces.
Why a digital meter/thermometer?	Higher resolution, no parallax, no interpolation, so a lower percentage uncertainty.
Why monochromatic light?	A single wavelength; white light has many wavelengths that refract or diffract by different angles, so the angle is uncertain.
Why a fixed resistor in series?	To limit the maximum current, preventing overheating, a short circuit or damage to the ammeter/supply (e.g. when the diode conducts).
Why start the variable resistor at maximum?	This gives the minimum current, so there is no overheating or damage to components.
A higher-power lamp is worse	It heats the sample (solution expands or evaporates, the solar cell warms), changing the property being measured.
Effect of air resistance or friction	It reduces the velocity, so r , D or the range is smaller, so the calculated value (e.g. e , v) is too small.
Control variable	Name it (not just "temperature of the room") and give the method: "same volume of oil, checked with a measuring cylinder each time"; "same release height, marked on the track".

10. Final checklist before the exam

Can you, without notes...	Can you, without notes...
State the resolution of every instrument in section 2 and choose between them with a reason?	Find % uncertainty from half resolution and from half range, with correct rounding?
Compare a result with a table value by both the limit method and the % difference method?	Spot four different faults in a results table?
Plot a graph that earns all the marks (labels, 1-2-5 scale, $\geq$ half grid, ± 1 mm, best fit)?	Linearise any equation in section 5 and say what the gradient and intercept give?
Give a hazard + control pair for masses, wires, lasers, hot water and sound?	Distinguish random and systematic error and say how each is reduced?
Draw the circuits for I - V , resistivity, e.m.f./internal resistance and a solar cell?	Describe each practical in section 7 with measurements, technique and analysis?